Chapter 4, Problem 4RQ

Problem

Refer to Fig. 4.67. The Thevenin resistance at terminals *a* and *b* is:

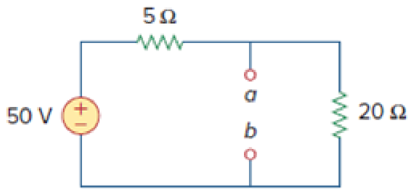
(a) 25 Ω

(b) 20 Ω

(c) 5 Ω

(d) 4 Ω

Figure. 4.67:



Step-by-step solution

Step 1 of 2

Find the Thevenin resistance at terminals *a* and *b* by short circuit the voltage source 50 V.

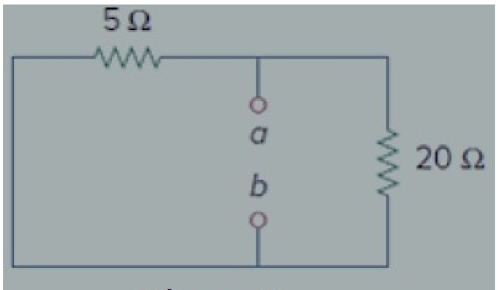


Figure 1

Step 2 of 2

The two resistors $20\ \Omega$ and $5\ \Omega$ are in parallel.

$$\begin{aligned} R_{\text{Th}} &= \frac{20\times 5}{20+5} \\ &= 4\ \Omega \end{aligned}$$

Thus, Thevenin resistance at terminals *a* and *b* is $\boxed{4\ \Omega}$.

Correct option is $\boxed{\text{d}}$.